

Surya GS Chitti

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Education

BITS Pilani

2020–2025

M.Sc. Mathematics + B.E. Civil Engineering — Minor: Computing & Intelligence

ML, AI, Probability & Statistics, Optimization, Numerical Methods, Graph Theory

Technical Skills

Core: Deep Learning, Self-Supervised Learning, Knowledge Distillation, Reinforcement Learning, Multi-Modal Fusion

Stack: Python, PyTorch, OpenCV, Docker, SQL, Linux, Git, PyTorch DDP

Experience

Mercedes-Benz Research & Development India — *ML Research Intern*

Sept 2025 – Present

End-to-End Perception & Planning, Applied Research Team

- **Data Pipeline:** Built a 5-stage pipeline from scratch (structure validation, per-frame quality gating, cross-sensor sync verification, road geometry classification, training-ready formatting) for **24 TB** of multi-sensor data (6 cameras, LiDAR, IMU). Also built a web app for the team to browse and curate data. Processing time per rosbag dropped from 225 to 25 minutes (**9×**).
- **Scenario Mining:** Training data was heavily imbalanced toward straight-road cruising. Built a clustering pipeline over **2.3M frames** to categorize recordings by maneuver type (lane changes, braking, turns, cruising) and rebalance the training distribution.
- **ML Infrastructure:** Built a config-driven training framework with automated post-training validation, checkpointing, and result visualization. All architecture experiments in the team ran on this infrastructure.
- **Architecture Ablation:** Systematically ablated temporal context (past 3 frames), auxiliary sensor fusion, and ego-motion conditioning. Scaled from single-camera to **multi-camera fusion** using learned spatial projection and self-supervised depth, removing the need for manual depth labels.
- **Knowledge Distillation:** Designed a 3-stage progressive distillation pipeline to transfer from a sensor-rich teacher to an image-only student. Each stage recovered more of the teacher's L2 waypoint accuracy: feature-level **31%**, +output-level **67%**, +ground-truth supervision **79%**.
- **Sequential Prediction:** Designed an autoregressive GRU decoder with relative deltas instead of independent MLP heads. L2 displacement error dropped by **23%**.
- **Sim-to-Real:** Implemented and compared **3 domain-adaptation approaches**: neural rendering, image-to-image translation, and direct synthetic training. Built the closed-loop evaluation framework to measure performance KPIs.
- **Sensor Calibration:** Built LiDAR-Camera and LiDAR-IMU calibration pipeline by implementing PnP-based extrinsics and hand-eye solver with online refinement via Powell optimization on alignment objectives.

AI & Robotics Lab, IISc Bangalore — *Research Intern*

Jan 2025 – Jul 2025

Mentored by [Dr. Rajini Makam](#) & [Prof. Suresh Sundaram](#) [Code] [Thesis]

- Built velocity estimation pipeline using Memory Neuron Networks for DRDO, robust under sensor degradation; **0.05 m/s RMSE** across **13 test trajectories**. Sole author of delivered production codebase.
- Audited pretrained depth model, traced failures to domain-specific preprocessing assumptions, re-derived for target domain. Evaluated visual navigation agents in simulation; identified training-distribution bias that fine-tuning could not override.

GVCL, IIIT Bangalore — *Research Intern*

Jul 2024 – Dec 2024

Mentored by [Prof. Jaya Sreevalsan Nair](#) [Thesis]

- Evaluated 3D point cloud segmentation models (KPConv, PointNet++) on multi-modal data. Built geometric alignment pipeline for cross-modal registration; applied ICP to merge sequential scans for spatial densification.

AI4Bharat, IIT Madras — *Research Intern*

Aug 2023 – Sep 2023

Mentored by [Md Safi Ur Rahman Khan](#) & [Prof. Mitesh Khapra](#)

- Processed multilingual NLP corpora (**2,000+ videos**, **3,000+ websites**) with script-aware filtering for low-resource language translation. Built prompt collection interface for instruction tuning of large language models.

Technical Writing

- **Why Gradient Descent Works (And When It Doesn't):** Proved when simultaneous parameter updates backfire. Compared Newton's method and coordinate descent with derivations and interactive notebook.
- **Embeddings From First Principles:** BPE tokenization, Word2Vec (Skip-gram & CBOW) with full gradient derivations, 6 ablation experiments implemented from scratch.